

QUESTION-7

Single linked list using insertion in at any point

main.c	Output
<pre>1 #include <stdio.h> 2 #include <stdlib.h> 3 struct node { 4 int data; 5 struct node *next; 6 }; 7 struct node *head = NULL; 8 void insertBegin(int value) { 9 struct node *newNode = (struct node 10 *)malloc(sizeof(struct node)); 11 newNode->data = value; 12 newNode->next = head; 13 head = newNode; 14 } 15 void insertEnd(int value) { 16 struct node *newNode = (struct node 17 *)malloc(sizeof(struct node)); 18 struct node *temp = head; 19 newNode->data = value; 20 newNode->next = NULL; 21 if (head == NULL) { 22 head = newNode; 23 return; 24 } 25 while (temp->next != NULL) 26 temp = temp->next; 27 temp->next = newNode; 28 } 29 void insertPosition(int value, int pos) { 30 struct node *newNode, *temp; 31 int i; 32 newNode = (struct node *)malloc(sizeof 33 (struct node)); 34 newNode->data = value; 35 if (pos == 1) { 36 newNode->next = head; 37 head = newNode; 38 return; 39 } 40 temp = head; 41 for (i = 1; i < pos - 1 && temp != NULL; i++) 42 temp = temp->next; 43 if (temp == NULL) { 44 printf("Invalid Position\n"); 45 return; 46 } 47 newNode->next = temp->next; 48 temp->next = newNode; 49 } 50 void display() { 51 struct node *temp = head; 52 while (temp != NULL) { 53 printf("%d -> ", temp->data); 54 temp = temp->next; 55 } 56 printf("NULL\n"); 57 } 58 int main() { 59 insertBegin(10); 60 insertBegin(5); 61 insertEnd(20); 62 insertEnd(30); 63 insertPosition(15, 3); 64 display(); 65 return 0; 66 }</pre>	<pre>Linked List: 5 -> 10 -> 15 -> 20 -> 30 -> NULL --- Code Execution Successful ---</pre>
<pre>23 while (temp->next != NULL) 24 temp = temp->next; 25 temp->next = newNode; 26 } 27 void insertPosition(int value, int pos) { 28 struct node *newNode, *temp; 29 int i; 30 newNode = (struct node *)malloc(sizeof 31 (struct node)); 32 newNode->data = value; 33 if (pos == 1) { 34 newNode->next = head; 35 head = newNode; 36 return; 37 } 38 temp = head; 39 for (i = 1; i < pos - 1 && temp != NULL; i++) 40 temp = temp->next; 41 if (temp == NULL) { 42 printf("Invalid Position\n"); 43 return; 44 } 45 newNode->next = temp->next; 46 temp->next = newNode; 47 } 48 void display() { 49 struct node *temp = head; 50 while (temp != NULL) { 51 printf("%d -> ", temp->data); 52 temp = temp->next; 53 } 54 printf("NULL\n"); 55 }</pre>	<pre>Linked List: 5 -> 10 -> 15 -> 20 -> 30 -> NULL --- Code Execution Successful ---</pre>
<pre>57 return; 58 } 59 printf("Linked List: "); 60 while (temp != NULL) { 61 printf("%d -> ", temp->data); 62 temp = temp->next; 63 } 64 printf("NULL\n"); 65 } 66 int main() { 67 insertBegin(10); 68 insertBegin(5); 69 insertEnd(20); 70 insertEnd(30); 71 insertPosition(15, 3); 72 display(); 73 return 0; 74 }</pre>	<pre>Linked List: 5 -> 10 -> 15 -> 20 -> 30 -> NULL --- Code Execution Successful ---</pre>